RAM VAMSHI KRISHNA RUDRARAM

Data Engineer | Analytics Engineer | ETL Developer

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PROFESSIONAL SUMMARY

Data Engineer with nearly 2 years of professional experience designing scalable data pipelines and analytics solutions in the healthcare and life sciences domain. Skilled in Python, SQL, PySpark, and Apache Spark for batch and real-time data processing, with strong hands-on experience building ETL/ELT pipelines on AWS (EMR, Glue, S3, Redshift). Proven ability to design dimensional data models, build automated data quality frameworks, and deliver curated, analytics-ready datasets that power BI dashboards, regulatory reporting, and machine learning workflows. Experienced in workflow orchestration with Apache Airflow, query and warehouse performance tuning, and translating business requirements into clean, well-modeled datasets and actionable insights.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, PL/SQL, Shell Scripting, Scala (basic), Java (basic)

Big Data & Distributed Processing: Apache Spark, PySpark, Spark SQL, Spark Streaming, Hadoop, Hive, HDFS

AWS Cloud: S3, EMR, Glue, Lambda, Redshift, Athena, Kinesis, Step Functions, CloudWatch, IAM

Azure Cloud: Azure Data Factory (ADF), Azure Databricks, Synapse Analytics, ADLS Gen2, Blob Storage, Azure Functions, Azure Monitor

Data Warehousing: Snowflake, Amazon Redshift, Azure Synapse, BigQuery (familiar), Star & Snowflake Schemas, Dimensional Modeling, SCD Type 1/2

Workflow Orchestration: Apache Airflow, AWS Step Functions, Azure Data Factory, Control-M (basic)

Streaming & Messaging: Apache Kafka, AWS Kinesis, Spark Structured Streaming, Event Hubs

Databases: PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, DynamoDB

BI & Analytics: Power BI, Tableau, Looker (familiar), Excel (advanced), Data Visualization, Reporting

Data Formats: JSON, XML, Parquet, Avro, ORC, CSV, Delta Lake

DevOps & CI/CD: Git, GitHub, Bitbucket, Jenkins, Azure DevOps, Docker, Terraform (basic), Linux/Unix

Data Engineering Concepts: ETL/ELT, Data Modeling, Data Quality, Data Governance, Source-to-Target Mapping, Data Lineage, Metadata Management, Master Data Management

Analytics & ML Support: Feature Engineering, Data Preparation for ML, Pandas, NumPy, Jupyter, Statistical Analysis

Methodologies: Agile/Scrum, Kanban, TDD, Code Reviews, Sprint Planning

PROFESSIONAL EXPERIENCE

Data Engineer | Johnson & Johnson **Feb 2022 – Dec 2023**

Hyderabad, India | Healthcare / Life Sciences

- Developed scalable ETL pipelines using Python, PySpark, and SQL on AWS (EMR, Glue, S3, Redshift) to process clinical trial and patient healthcare datasets exceeding 5TB, enabling timely insights for research teams.
- Built batch and streaming pipelines with Apache Spark, Kafka, and AWS Kinesis for ingestion and transformation of high-volume HL7 and FHIR healthcare data, supporting near real-time analytics.
- Designed and maintained 25+ Apache Airflow DAGs for workflow orchestration, improving pipeline reliability and observability while reducing job failures by 35%.
- Modeled and built star-schema data marts in Amazon Redshift supporting clinical reporting, regulatory compliance, and operational dashboards consumed by 100+ analysts.
- Tuned complex SQL queries and optimized Redshift table design (distribution keys, sort keys, compression), reducing query execution time on multi-billion-row tables by 50%.
- Engineered data validation frameworks in Python to verify source-to-target mappings and transformations, catching data quality issues early and improving downstream data trust.
- Built reusable Python utilities and libraries for ingestion, transformation, logging, and validation, accelerating new pipeline development by 30% across the team.

- Migrated legacy on-premise ETL workloads to AWS, leveraging S3 for storage, Glue Catalog for metadata, and EMR/Spark for distributed processing, reducing infrastructure costs by approximately 30%.
- Prepared analytics-ready datasets for data scientists and biostatisticians, performing feature engineering, aggregation, and pivoting using Python (Pandas, NumPy) and Spark.
- Collaborated with BI teams to deliver curated datasets feeding Tableau and Power BI dashboards used by clinical operations and executive leadership.
- Worked in Agile/Scrum sprints, participating in sprint planning, code reviews, retrospectives, and stakeholder demos to deliver features iteratively.

EDUCATION

Master of Science in Computer Science – Missouri State University, Springfield, MO | Dec 2025

Bachelor of Technology in Computer Science – Koneru Lakshmaiah Education Foundation, India | May 2023